MIL-P-46122B(MR)
5 June 1975
SUPERSEDING
MIL-P-46122A(MR)
17 June 1970

# MILITARY SPECIFICATION

## PLASTIC MOLDING AND EXTRUSION MATERIAL, POLYVINYLIDENE FLUORIDE POLYMER AND COPOLYMER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of polyvinylidene fluoride molding and extrusion material.

1.2 Classification. The polyvinylidene fluoride material covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Homopolymer resin
  Grade A - Non electrical applications
  Grade E - Electrical applications

Type II - Copolymer resin
  Grade A - Non electrical applications
  Grade E - Electrical applications

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-gallon

FSC 9330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial power Frequencies.

D 150 - AC Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.

D 256 - Impact Resistance of Plastics and Electrical Insulating Materials

D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing

D 638 - Tensile Properties of Plastics

D 790 - Flexural Properties of Plastics

D 792 - Specific Gravity and Density of Plastics by Displacement

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.



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Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

3. REQUIREMENTS

3.1 Material. The material shall consist of virgin polyvinylidene fluoride homopolyer or copolymer, without plasticizer. Type I material shall consist of the homopolymer. Type II material shall be a copolymer predominately polyvinylidene fluoride and containing polytetrafluoroethylene.

3.2 Property values. The material shall conform to the property values specified in table I when tested as specified in the applicable procedure of 4.3. When grade E is specified by the procuring activity (see 6.2), the electrical properties specified in table II shall be included as requirements.

Table I. Property values (excluding electrical)

| Property | Value | |
|---|---|---|
| | Type I Grades A and E | Type II Grades A and E |
| Tensile strength, psi, min. | 5500 | 5500 |
| Elongation, percent, min. | 50 | 100 |
| Flexural modulus, psi, min. | 200,000 | 195,000 |
| Impact strength, Izod, ft-lb per inch of notch, min. | 3.8 | 2.3 |
| Specific gravity, min. | 1.75 | 1.76 |
| Specific gravity, max. | 1.78 | 1.79 |
| Melting point, °C | 167-170 | 150-155 |
| Melting point, °F | 332-338 | 302-311 |



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Table II. Electrical property values

| Property | Type I Grade E | Type II Grade E |
|---|---|---|
| Dielectric constant, max. | | |
| at 60 Hz | 8.4 | 8.1 |
| $10^6$ Hz | 6.4 | 6.1 |
| Dissipation factor, max. | | |
| at 60 Hz | 0.049 | 0.036 |
| $10^6$ Hz | 0.159 | 0.143 |
| Dielectric strength in air, volts per mil, min. | 1280 | 1280 |

3.3 Color. Unless otherwise specified by the procuring activity (see 6.2) the color shall be natural, which is off white to light tan in sections above 0.025 inches thick. Thinner sections shall be transparent to semi-transparent.

3.4 Workmanship. Unless otherwise specified by the procuring activity (see 6.2) the material shall be in the form of pellets. The pellets shall be uniform in form and color as determined visually. The uniformity shall be within each container and from container to container. The material shall be clean and free from foreign material as determined by visual examination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type submitted for delivery at one time.



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4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in color, and workmanship. The sample unit for this examination specified in table III shall be one container of any size to be purchased.

Table III. Examination of the material for defects in appearance and workmanship

| Examine | Defect |
|---|---|
| Color | Color not natural, unless otherwise specified |
| Workmanship | Form not uniform |
| | Color not uniform |
| | Not clean, presence of foreign material |

4.2.2.1.2 Examination of the preparation for delivery. An examination shall be made in accordance with table IV, to determine that packaging, packing and marking comply with specified contract requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. All shipping containers fully prepared for delivery shall be examined for closure defects.



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Table IV. Examination for preparation for delivery

| Examine | Defect |
|---|---|
| Packing | Not level specified; not in accordance with contract requirements |
| | Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability |
| | Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers |
| Quantity of material | Less than specified or indicated quantity |
| Weight | Gross weight exceeds specified requirements |
| Markings | Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location or sequence or method of application |
| | Not in accordance with contract requirements |

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

| Examination paragraph | Inspection level | AQL |
|---|---|---|
| 4.2.2.1.1 | I | 1.0 |
| 4.2.2.1.2 | S-2 | 2.5 |

4.2.2.2 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. They shall be made on each lot of material offered against this specification and, in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.



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4.2.3 Testing. The material shall be tested for the characteristics listed in table I, and table II (when applicable), in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 100 pounds. The sample unit shall consist of sufficient material to prepare all required specimens plus the sample for flow rate determination. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be the averaged results of the specimens, unless only one specimen or determination is specified for testing.

4.3 Test methods.

4.3.1 Specimen preparation. Unless otherwise specified, all specimens shall be prepared by injection molding under conditions specified by the manufacturer.

4.3.2 Specimen conditioning and testing. All specimens shall be conditioned in accordance with procedure A of ASTM D 618. Testing shall be at 23° ± 2°C (73.4° ± 6°F) and 50 ± 5 percent humidity except for the melting point determination.

4.3.3 Tensile strength and elongation. Three specimens shall be tested in accordance with ASTM D 638, using testing speed B. Specimens shall be 1/8 inch thick and dimensions and tolerances shall conform to type I of figure 1 of ASTM D 638.

4.3.4. Flexural modulus. Five specimens shall be tested in accordance with ASTM D 790, using procedure A. Specimens shall be 1/4 inch thick, and other dimensions and crosshead speed shall conform to table I of ASTM D 790. The tangent method shall be used for calculations.

4.3.5 Impact strength. Five specimens shall be tested for Izod impact strength in accordance with ASTM D 256 using method A. Specimens shall be 1/4 inch thick and shall conform to figure 2 of ASTM D 256.

4.3.6 Specific gravity. Two specimens shall be tested in accordance with ASTM D 792, using method A-1 or A-2. Specimens for specific gravity determination may be obtained from those used to determine tensile strength, flexural modulus, or impact strength.

4.3.7 Melting point. One shaving, 10 to 15 mils thick, shall be tested using a Fisher-Johns melting point apparatus, or equal. The shaving may be obtained from specimens used to determine tensile strength, flexural modulus, or impact strength. The melting point apparatus shall be used in accordance with instructions furnished by the manufacturer of the instrument.



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4.3.8 Dielectric constant and dissipation factor. The dielectric constant and dissipation factor shall be determined on 3 specimens, each 4 inches in diameter and 1/8 inch thick, in accordance with ASTM D 150. Testing shall be at 60 Hz and $10^6$ Hz.

4.3.9 Dielectric strength. The dielectric strength of 3 specimens each 0.005 inch thick, shall be determined by the "short-time test" of ASTM D 149. Testing shall be in air using 1 inch electrodes. The specimen shall be either 0.005 inch thick film or 0.005 inch thick wire insulation.

5. PREPARATION FOR DELIVERY

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), the material in the quantity specified (see 6.2), shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, containers shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified (see 6.2), the material, in the quantity specified (see 6.2), shall be in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, containers shall be of uniform shape and size with minimum cube tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.



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5.1.3 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.2 Marking. In addition to any special marking specified in the contract or order (see 6.2) containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Polyvinylidene fluoride material is intended for use in seals, gaskets, poppets and similar items where a thermoplastic material which may be useable in the temperature range from -62°C (-80°F) to 138°C (280°F) is required. The material is intended also, for use in long life exterior siding and building components because of the relatively good resistance to abrasion and chemical attack as well as stability to heat and ultraviolet light. Other applications include wire cable insulation, electronic and electrical components, chemical process equipment components, and aerospace components in contact with storable propellants. Type I material is used in applications requiring greater flexural modulus or melting point. Type II material is used in applications requiring greater flexibility and elongation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and grade of material required (see 1.2, 3.1, and 6.1).
(c) Color other than natural (see 3.3), or form other than pellets (see 3.4, if required).
(d) Level of packing required (see 5.1).
(e) If other containers are required (see 5.1.1 and 5.1.2).
(f) Quantity in shipping container (see 5.1.1 and 5.1.2).
(g) Special markings required (see 5.2).



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6.3 Cross index for material. Type I material covered by this specification corresponds to the material covered by MIL-P-46122(MR). Because of technological changes, properties may not be exactly equivalent.

Custodian:
Army - MR

Review activities:
Army - EL, GI
DSA-GS

User activities:
Army - AV, MI, AT

Preparing activity:
Army - MR

Project No. 9330-A509



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION **MIL-P-46122B(MR), Plastic Molding and Extrusion Material, Polyvinyldene Fluoride Polymer and Copolymer**

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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